



VP AIR TEST NEET (2025-26)

PART TEST - 01

DROPPER

DURATION: 180 Minutes

Date: 24-08-2025

M. MARKS: 720

ANSWER KEY

1. (3)	46. (3)	91. (1)	136. (2)
2. (2)	47. (3)	92. (3)	137. (3)
3. (2)	48. (2)	93. (1)	138. (1)
4. (1)	49. (2)	94. (2)	139. (2)
5. (1)	50. (2)	95. (3)	140. (1)
6. (3)	51. (1)	96. (4)	141. (2)
7. (3)	52. (4)	97. (2)	142. (4)
8. (1)	53. (2)	98. (1)	143. (2)
9. (4)	54. (3)	99. (2)	144. (2)
10. (3)	55. (4)	100. (2)	145. (2)
11. (2)	56. (3)	101. (2)	146. (3)
12. (4)	57. (4)	102. (4)	147. (3)
13. (1)	58. (1)	103. (4)	148. (3)
14. (3)	59. (3)	104. (1)	149. (3)
15. (3)	60. (1)	105. (2)	150. (2)
16. (1)	61. (2)	106. (2)	151. (2)
17. (2)	62. (1)	107. (1)	152. (4)
18. (1)	63. (1)	108. (4)	153. (3)
19. (4)	64. (3)	109. (2)	154. (3)
20. (2)	65. (3)	110. (3)	155. (2)
21. (1)	66. (4)	111. (2)	156. (4)
22. (1)	67. (2)	112. (3)	157. (4)
23. (3)	68. (3)	113. (3)	158. (2)
24. (4)	69. (4)	114. (3)	159. (2)
25. (3)	70. (3)	115. (4)	160. (4)
26. (2)	71. (1)	116. (1)	161. (2)
27. (2)	72. (2)	117. (2)	162. (1)
28. (1)	73. (3)	118. (4)	163. (4)
29. (4)	74. (3)	119. (3)	164. (3)
30. (1)	75. (1)	120. (2)	165. (1)
31. (2)	76. (1)	121. (3)	166. (2)
32. (4)	77. (3)	122. (4)	167. (4)
33. (3)	78. (2)	123. (2)	168. (1)
34. (3)	79. (1)	124. (4)	169. (3)
35. (1)	80. (4)	125. (1)	170. (2)
36. (3)	81. (1)	126. (1)	171. (1)
37. (4)	82. (4)	127. (4)	172. (3)
38. (4)	83. (2)	128. (2)	173. (4)
39. (4)	84. (1)	129. (3)	174. (2)
40. (2)	85. (2)	130. (3)	175. (2)
41. (2)	86. (1)	131. (1)	176. (3)
42. (3)	87. (3)	132. (4)	177. (4)
43. (2)	88. (1)	133. (3)	178. (3)
44. (1)	89. (1)	134. (2)	179. (2)
45. (2)	90. (2)	135. (3)	180. (3)

HINTS & SOLUTIONS**1. (3)**

$$[\mu_0] = [c]^a [G]^b [h]^c [k]^d [e]^f$$

$$[\mu_0] = [M L T^{-2} A^{-2}]$$

$$[c] = [L T^{-1}]$$

$$[G] = [M^{-1} L^3 T^{-2}]$$

$$[h] = [M L^2 T^{-1}]$$

$$[k] = [M L^2 T^{-2} \theta^{-1}]$$

$$[e] = [AT]$$

By equating the powers for each base dimension on both sides,

$$\text{For } M: 1 = -b + c + d \quad \dots(1)$$

$$\text{For } L: 1 = a + 3b + 2c + 2d \quad \dots(2)$$

$$\text{For } T: -2 = -a - 2b - c - 2d + f \quad \dots(3)$$

$$\text{For } A: -2 = f \quad \dots(4)$$

$$\text{For } \theta: 0 = -d \quad \dots(5)$$

Solving this system gives a unique solution:

From (5), we find $d = 0$.

From (4), we find $f = -2$.

Substituting $d = 0$ into (1) gives $1 = -b + c$.

Substituting $d = 0$ and $f = -2$ into (3) simplifies it to $0 = -a - 2b - c$.

Substituting $d = 0$ into (2) gives $1 = a + 3b + 2c$.

Solving the remaining system for a , b , and c yields $a = -1$, $b = 0$, and $c = 1$.

Therefore, the exponents are $a = -1$, $b = 0$, $c = 1$, $d = 0$, and $f = -2$.

This results in the dimensional formula

$$[c^{-1} G^0 h^1 k^0 e^{-2}].$$

(NCERT Ed. 2025-26, 11th, Page No. 9)

2. (2)

The term αt must be dimensionless,

dimension of α is $[T^{-1}]$. The term $\frac{\beta}{\alpha} (1 - e^{(-\alpha t)})$

must also have the same dimensions as P as the term $(1 - e^{(-\alpha t)})$ is dimensionless. Therefore, the dimension of β/α must be the same as the dimension of P .

$$\text{So, } [\beta]/[\alpha] = [P].$$

$$[\beta]/[T^{-1}] = [P],$$

$$[\beta] = [P][T^{-1}].$$

(NCERT Ed. 2025-26, 11th, Page No. 8)

3. (2)

(A) 9.999 has 3 decimal places.

$$10.0001 - 9.999 = 0.0011.$$

Rounding to 3 decimal places gives 0.001. This result has only 1 significant figure.

(B) $(5.5 \times 10^{-4}) + (2.22 \times 10^{-6})$: For addition with scientific notation, the exponents must be made equal first.

$$= (550 \times 10^{-6}) + (2.22 \times 10^{-6}).$$

$$= (550 + 2.22) \times 10^{-6} = 552.22 \times 10^{-6}$$

Rounding this to the tens place gives 550.

The result is 550×10^{-6} , which is 5.5×10^{-4} in proper scientific notation.

This has 2 significant figures.

(C) $(2.000)^3$: The base 2.000 has 4 significant figures.

$$(2.000) \times (2.000) \times (2.000) = 8.000.$$

The result must be reported with 4 significant figures, which is 8.000.

(D) 3.00×1.50 :

3.00 has 3 significant figures. 1.50 has 3 significant figures.

The result must be reported with 3 significant figures.

$3.00 \times 1.50 = 4.50$. This answer already has 3 significant figures.

(NCERT Ed. 2025-26, 11th, Page No. 5)

4. (1)

$$S = \log_{10} \left[\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{20^2}\right) \right]$$

Factor each term using difference of squares,

$$1 - \frac{1}{k^2} = \frac{(k-1)(k+1)}{k^2}:$$

$$S = \log_{10} \left[\left(\frac{1 \cdot 3}{2 \cdot 2}\right) \left(\frac{2 \cdot 4}{3 \cdot 3}\right) \left(\frac{3 \cdot 5}{4 \cdot 4}\right) \dots \left(\frac{19 \cdot 21}{20 \cdot 20}\right) \right]$$

$$S = \log_{10} \left(\frac{1}{2} \times \frac{21}{20} \right)$$

Final result:

$$S = \log_{10} \left(\frac{21}{40} \right) = \log_{10}(21) - \log_{10}(40)$$

5. (1)

By similar triangles, the water's radius r and height h are related by $\frac{r}{h} = \frac{5}{10}$, so $r = \frac{h}{2}$.

$$\text{Surface Area: } A = \pi r^2 = \pi \left(\frac{h}{2} \right)^2 = \frac{\pi}{4} h^2$$

$$\text{Volume: } V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi \left(\frac{h}{2} \right)^2 h = \frac{\pi}{12} h^3$$

$$\frac{dV}{dt} = \frac{d}{dt} \left(\frac{\pi}{12} h^3 \right) = \frac{\pi}{4} h^2 \frac{dh}{dt}$$

$$\text{We are given: } \frac{dV}{dt} = -\frac{\pi}{4} \sqrt{h}$$

$$\text{So, } \frac{\pi}{4} h^2 \frac{dh}{dt} = -\frac{\pi}{4} \sqrt{h}, \quad \frac{dh}{dt} = \frac{-\sqrt{h}}{h^2} = -h^{-3/2}$$

$$\frac{dA}{dt} = \frac{d}{dt} \left(\frac{\pi}{4} h^2 \right) = \frac{\pi}{2} h \frac{dh}{dt}$$

At the specific instant $h = 4$:

$$\frac{dh}{dt} = -(4)^{-3/2} = -\frac{1}{4^{3/2}} = -\frac{1}{8} \text{ m/s}$$

Substitute into the equation for $\frac{dA}{dt}$:

$$\frac{dA}{dt} = \frac{\pi}{2} (4) \left(-\frac{1}{8} \right) = -\frac{\pi}{4} \text{ m}^2/\text{s}$$

6. (3)

The measurement 2.321 cm has three decimal places, meaning it is precise to 0.001 cm, which is 0.01 mm. An instrument must have a least count of at least 0.01 mm to make this measurement.

LC of A: $\text{LC} = 1 \text{ MSD} - 1 \text{ VSD} = 1 \text{ mm} - (19/20) \text{ mm} = 1/20 \text{ mm} = 0.05 \text{ mm}$.

LC of B: $\text{LC} = 1 \text{ MSD} - 1 \text{ VSD} = 0.5 \text{ mm} - (24/25) \times 0.5 \text{ mm} = (1/25) \times 0.5 \text{ mm} = 0.02 \text{ mm}$.

LC of C: $\text{LC} = \text{Pitch} / \text{Divisions} = 0.5 \text{ mm} / 50 = 0.01 \text{ mm}$.

LC of D: $\text{LC} = \text{Pitch} / \text{Divisions} = 1 \text{ mm} / 50 = 0.02 \text{ mm}$.

Only Instrument C has a least count (0.01 mm) that is fine enough to verify the measurement to the required precision.

(NCERT Ed. 2025-26, 11th, Page No. 3)

7. (3)

Let $P = \cos(\pi/7) \times \cos(2\pi/7) \times \cos(4\pi/7)$.

Multiply and divide the entire expression by $2\sin(\pi/7)$.

$$P = \frac{2\sin(\pi/7)\cos(\pi/7)\cos(2\pi/7)\cos(4\pi/7)}{2\sin(\pi/7)}$$

Notice that $2\sin(\pi/7)\cos(\pi/7) = \sin(2 \times \pi/7) = \sin(2\pi/7)$.

$$P = \frac{\sin(2\pi/7)\cos(2\pi/7)\cos(4\pi/7)}{2\sin(\pi/7)}$$

Multiply the numerator and denominator by 2.

$$P = \frac{2\sin(2\pi/7)\cos(2\pi/7)\cos(4\pi/7)}{2 \cdot 2\sin(\pi/7)}$$

$$P = \frac{\sin(4\pi/7)\cos(4\pi/7)}{4\sin(\pi/7)}$$

Multiplying the numerator and denominator by 2:

$$P = \frac{2\sin(4\pi/7)\cos(4\pi/7)}{2 \cdot 4\sin(\pi/7)} \quad P = \frac{\sin(8\pi/7)}{8\sin(\pi/7)}$$

Use the identity $\sin(\pi + \theta) = -\sin(\theta)$.

$$\sin(8\pi/7) = \sin(\pi + \pi/7) = -\sin(\pi/7)$$

$$P = \frac{-\sin(\pi/7)}{8\sin(\pi/7)}$$

$$\Rightarrow P = -\frac{1}{8}$$

8. (1)

The method of reducing random error is by taking more and more number of observations.

(NCERT Ed. 2025-26, 11th, Page No. 3)

9. (4)

$$\begin{aligned} & \frac{d}{dx} \left(x + \frac{1}{x} + \log x + \tan x \right) \\ &= 1 - \frac{1}{x^2} + \frac{1}{x} + \sec^2 x \end{aligned}$$

10. (3)

If x is the length of each side of an equilateral triangle and A is its area, then

$$A = \frac{\sqrt{3}}{4} x^2$$

$$\Rightarrow \frac{dA}{dt} = \frac{\sqrt{3}}{4} 2x \frac{dx}{dt}$$

Here, $x = 10 \text{ cm}$

and $\frac{dx}{dt} = 2 \text{ cm/sec}$

$$\Rightarrow \frac{dA}{dt} = 10\sqrt{3} \text{ cm}^2/\text{sec}$$

11. (2)

$$\text{Area} = \pi ab$$

Divide the entire equation by 400:

$$\frac{25x^2}{400} + \frac{16y^2}{400} = \frac{400}{400}$$

$$\frac{x^2}{16} + \frac{y^2}{25} = 1$$

Identify a^2 and b^2 :

$$a = \sqrt{16} = 4$$

$$b = \sqrt{25} = 5$$

$$\text{Area} = \pi \times 4 \times 5 = 20\pi$$

The total area of the elliptical logo element is $20\pi \text{ cm}^2$.

12. (4)

Boltzmann Constant (k_B): $[ML^2T^{-2}K^{-1}]$

Pressure (P): $[ML^{-1}T^{-2}]$

Mass (m): $[M]$

Mean Free Path (λ): $[L]$

Thermal Conductivity (κ): $[MLT^{-3}K^{-1}]$

Form the Dimensional Equation

$$[M^1L^2T^{-3}K^{-1}] = [ML^2T^{-2}K^{-1}]^a \times [ML^{-1}T^{-2}]^b \times [M]^c \times [L]^d$$

$$(M^1L^2T^{-3}K^{-1}) = M^{(a+b+c)} \times L^{(2a-b+d)} \times T^{(-2a-2b)} \times K^{-a}$$

Equating Powers

For K : $-1 = -a \Rightarrow a = 1$

For T : $-3 = -2a - 2b$

$$\Rightarrow -3 = -2(1) - 2b \Rightarrow b = \frac{1}{2}$$

$$\text{For } M: 1 = a + b + c \Rightarrow 1 = 1 + \frac{1}{2} + c \Rightarrow c = -\frac{1}{2}$$

$$\text{For } L: 1 = 2a - b + d \Rightarrow 1 = 2(1) - \frac{1}{2} + d \Rightarrow d = -\frac{1}{2}$$

Step 3: Final Expression

$$K = C \cdot k_B^1 \cdot P^{1/2} \cdot m^{-1/2} \cdot \lambda^{-1/2}$$

$$K = C k_B \sqrt{\frac{P}{m\lambda}}$$

(NCERT Ed. 2025-26, 11th, Page No. 9)

13. (1)

$$(A) \sin 37^\circ = \frac{3}{5} - II$$

$$(B) \cos 127^\circ = \cos(180^\circ - 53^\circ) = -\cos 53^\circ = -\frac{3}{5} - I$$

$$(C) \tan 307^\circ = \tan(360^\circ - 53^\circ) = -\tan 53^\circ = -\frac{4}{3} - III$$

$$(D) \cos 323^\circ = \cos(360^\circ - 37^\circ) = \cos 37^\circ = \frac{4}{5} - IV$$

14. (3)

$$(1) v = \sqrt{\frac{k_B T}{m}}$$

The dimension of $k_B T$ (energy) is $[ML^2T^{-2}]$.

The dimension of m (mass) is $[M]$.

The dimension of $\frac{k_B T}{m}$ is

$$[ML^2T^{-2}] / [M] = [L^2T^{-2}]$$

Taking the square root: $\sqrt{L^2T^{-2}} = [LT^{-1}]$.

This expression is dimensionally correct.

$$(2) v = \frac{S}{\eta}$$

The dimension of S (surface tension) is $[MT^{-2}]$.

The dimension of η (viscosity) is $[ML^{-1}T^{-1}]$.

The dimension of $\frac{S}{\eta}$ is

$$[MT^{-2}] / [ML^{-1}T^{-1}] = M^{1-1} L^{0-(-1)} T^{-2-(-1)} = [L^1T^{-1}]$$

This expression is dimensionally correct.

$$(3) v = \frac{E}{\rho}$$

The dimension of E (modulus of elasticity) is $[ML^{-1}T^{-2}]$.

The dimension of ρ (density) is $[ML^{-3}]$.

The dimension of $\frac{E}{\rho}$ is

$$[ML^{-1}T^{-2}] / [ML^{-3}] = M^{1-1} L^{-1-(-3)} T^{-2} = [L^2T^{-2}]$$

This expression has the dimensions of speed squared (v^2). This expression is dimensionally incorrect.

$$(4) v = \sqrt{\frac{E}{\rho}}$$

From part (c), $\frac{E}{\rho}$ has dimension $[L^2T^{-2}]$.

Taking the square root: $\sqrt{L^2T^{-2}} = [LT^{-1}]$.

This expression is dimensionally correct.

(NCERT Ed. 2025-26, 11th, Page No. 8)

15. (3)

Pressure Gradient (change in pressure per unit distance) and Force Density (force per unit volume) both have the dimensional formula $[ML^{-2}T^{-2}]$.

Latent Heat (energy per unit mass) and Gravitational Potential (work done per unit mass) both have the dimensional formula $[L^2T^{-2}]$.

Planck's Constant and Angular Momentum share the dimensional formula $[ML^2T^{-1}]$.

Boltzmann Constant and Heat Capacity (energy per unit temperature) share the dimensional formula $[ML^2T^{-2}K^{-1}]$.

(NCERT Ed. 2025-26, 11th, Page No. 7)

16. (1)
Both statements I and II are correct.
(NCERT Ed. 2025-26, 11th, Page No. 8)

17. (2)

$$\frac{\Delta R_{eq}}{R_{eq}} = R \left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2} \right)$$

$$\Rightarrow \frac{20 \times 5}{25} \left(\frac{0.1}{25} + \frac{0.4}{400} \right)$$

$$\Rightarrow 4 \left(\frac{0.1}{25} + \frac{0.1}{100} \right)$$

$$= 4 \left(\frac{0.5}{100} \right) = 0.02$$

$$\frac{\Delta R_{eq}}{R_{eq}} \times 100 = 2\%$$

(NCERT Ed. 2025-26, 11th, Page No. 6)

18. (1)

$$v = \pi r^2 h$$

$$\frac{\Delta v}{v} = \frac{2\Delta r}{r} + \frac{\Delta h}{h} = \frac{2 \times (0.1)}{10} + \frac{1}{1000}$$

$$\frac{\Delta v}{v} = \frac{2}{100} + \frac{1}{1000} = \frac{21}{1000}$$

% error = 2.1%

(NCERT Ed. 2025-26, 11th, Page No. 6)

19. (4)

$$A = 2 \int_{y=0}^{y=4} x dy = 2 \int_{y=0}^{y=4} \sqrt{y} dy = \frac{32}{3} m^2$$

20. (2)

$$\theta = \frac{3t^3}{10} - 2t$$

$$\frac{d\theta}{dt} = \frac{9t^2}{10} - 2$$

$$\frac{d^2\theta}{dt^2} = \frac{18t}{10}$$

$$\left(\frac{d^2\theta}{dt^2} \right)_{t=2} = 3.6 \text{ rad/s}^2$$

21. (1)

$$-\sqrt{8^2 + (-6)^2} \leq (8 \sin \theta - 6 \cos \theta) \leq \sqrt{8^2 + (-6)^2}$$

$$-10 \leq (8 \sin \theta - 6 \cos \theta) \leq 10$$

then greatest value = 10

22. (1)

$$\int_{-\pi/2}^{\pi/2} \sqrt{4 - (2 \sin \theta)^2} \times 2 \cos \theta d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \sqrt{4(1 - \sin^2 \theta)} \times 2 \cos \theta d\theta$$

$$= \int_{-\pi/2}^{\pi/2} 4 \cos^2 \theta d\theta = \int_{-\pi/2}^{\pi/2} 4 \left(\frac{1 + \cos 2\theta}{2} \right) d\theta$$

$$= 2 \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta) d\theta = 2\pi$$

23. (3)

Let $T_{\text{initial}} = 2\pi \sqrt{L_{\text{initial}} / g}$.

$$T_{\text{final}} = 2\pi \sqrt{L_{\text{initial}} (1 + \alpha \Delta T) / g}$$

$$T_{\text{final}} = \left[2\pi \sqrt{L_{\text{initial}} / g} \right] \cdot \sqrt{1 + \alpha \Delta T}$$

$$T_{\text{final}} = T_{\text{initial}} \cdot (1 + \alpha \Delta T)^{1/2}$$

Applying the approximation:

$$T_{\text{final}} \approx T_{\text{initial}} \cdot \left(1 + \frac{1}{2} \alpha \Delta T \right)$$

$$\text{Fractional Change} \approx \frac{T_{\text{initial}} \left(1 + \frac{1}{2} \alpha \Delta T \right) - T_{\text{initial}}}{T_{\text{initial}}}$$

$$\approx \frac{1}{2} \alpha \Delta T$$

$$\text{Fractional Change} \approx \frac{1}{2} \cdot (1.9 \times 10^{-5}) \cdot (10)$$

$$= 9.5 \times 10^{-5}$$

Total time lost in a day:

$$\text{Time Lost} \approx (9.5 \times 10^{-5}) \times 86400 \text{ seconds}$$

$$\text{Time Lost} \approx 8.208 \text{ seconds}$$

(NCERT Ed. 2025-26, 11th, Page No. 9)

24. (4)

Slope at x_0 is positive in graph (ii)

25. (3)

$$y = 3x^2 - 3$$

put $y = 0$

$$3(x^2 - 1) = 0$$

$$x = \pm 1$$

put $x = 0$

$$y = -3$$

26. (2)

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c$$

$$= 10 \left(\frac{1 \text{ kg}}{\alpha \text{ kg}} \right)^1 \left(\frac{1 \text{ m}}{\beta \text{ m}} \right)^2 \left(\frac{1 \text{ s}}{\delta \text{ s}} \right)^{-2}$$

$$= \frac{10\delta^2}{\alpha\beta^2} = 10\alpha^{-1}\beta^{-2}\delta^2$$

(NCERT Ed. 2025-26, 11th, Page No. 11)

27. (2)

When $z = 0$, $y = e^0 = 1$

Substitute $y = 1$ into the first equation:

$$x^2(1) = \ln(1) + \sin(x)$$

$$x^2 = \sin(x)$$

The only solution is $x = 0$.

So, the point is: $(x, y, z) = (0, 1, 0)$

Differentiate $x^2 y = \ln(y) + \sin(x)$ with respect to y :

$$\left(2x \cdot \frac{dx}{dy} \right) y + x^2 = \frac{1}{y} + \cos(x) \frac{dx}{dy}$$

$$\frac{dx}{dy} = \frac{\frac{1}{y} - x^2}{2xy - \cos(x)}$$

$$\text{At } (x = 0, y = 1): \frac{dx}{dy} = \frac{1/1 - 0^2}{0 - \cos(0)} = \frac{1}{-1} = -1$$

$$\text{Since } y = e^z: \quad \frac{dy}{dz} = e^z$$

$$\text{At } z = 0: \frac{dy}{dz} = e^0 = 1$$

$$\frac{dx}{dz} = \frac{dx}{dy} \cdot \frac{dy}{dz} = (-1)(1) = -1$$

28. (1)

$$\int \left[5 \sec^2(x) + (x^2 - 4x + 4) - x^2 + 4x - 5 \tan^2(x) \right] dx$$

$$= \int \left[(5 \sec^2(x) - 5 \tan^2(x)) + (x^2 - x^2) + (-4x + 4x) + 4 \right] dx$$

Use the identity $\sec^2(x) - \tan^2(x) = 1$. The expression becomes:

$$= \int [5(1) + 0 + 0 + 4] dx$$

$$= \int [5 + 4] dx$$

$$= \int 9 dx = 9x + C$$

29. (4)

Dimensional formula of $K = [ML^2T^{-2}]$

Dimensional formula of $m^2 a = [M^2LT^{-2}]$

Dimensional formula of $m^2 v^3 = [M^2L^3T^{-3}]$

(NCERT Ed. 2025-26, 11th, Page No. 9)

30. (1)

L.C. = 0.01 mm

Total reading = MSR + CSR

$$= (3 \text{ mm}) + 25 \times (\text{L.C.})$$

$$= (3 \text{ mm}) + 25 \times (0.01 \text{ mm}) = 3.25 \text{ mm}$$

(NCERT Ed. 2025-26, 11th, Page No. 11)

31. (2)

If the number of terms in the A.P. are n , then 10th term from the end will be $(n - 9)$ th term from starting.

$$\therefore 5 + (n - 10) \times 3 = 95$$

$$\Rightarrow n = 40.$$

32. (4)

Work and Torque both are different physical quantity but having same dimension and can't be added or subtract.

Only same physical quantity can be added or subtract.

(NCERT Ed. 2025-26, 11th, Page No. 8)

33. (3)

Ψ is energy per unit volume per unit time.

$$[\Psi] = \frac{[\text{Energy}]}{([\text{Volume}][\text{Time}])} = \frac{[ML^2T^{-2}]}{([L^3][T])} = [ML^{-1}T^{-3}]$$

Apply the Principle of Homogeneity:

$$[\Psi] = \left[\frac{(b \times v^2)}{x} \right]$$

$$[b] = \frac{([\Psi] \times [x])}{[v]^2}$$

$$[b] = \frac{([ML^{-1}T^{-3}] \times [L])}{[L^2T^{-2}]}$$

$$[b] = [ML^{-2}T^{-1}]$$

(NCERT Ed. 2025-26, 11th, Page No. 8)

34. (3)

$$a_m = \frac{2.29 + 2.24 + 2.34}{3} = 2.29$$

$$\Delta a_m = \frac{0 + 0.05 + 0.05}{3} = \frac{1}{30}$$

(NCERT Ed. 2025-26, 11th, Page No. 6)

35. (1)

$$(2.004)^3 = (2 + 0.004)^3$$

$$= 2^3 \left[1 + \frac{0.004}{2} \right]^3 = 8 \left[1 + \frac{0.004}{2} \right]^3$$

$$= 8 \left[1 + \frac{3 \times 0.004}{2} \right] = 8.048$$

36. (3)

Put $\frac{dy}{dx} = 0 \Rightarrow 10x - 2 = 0$

$$x = \frac{1}{5}$$

$$\frac{d^2y}{dx^2} = 10 > 0, \text{ hence } x = \frac{1}{5} \text{ is a point of minima.}$$

$$y_{\min} = 5 \left(\frac{1}{25} \right) - 2 \left(\frac{1}{5} \right) + 5 = \frac{24}{5}$$

37. (4)

% Error in mass (m):

$$(\Delta m / m) \times 100 = (0.02 / 40.00) \times 100 = 0.05\%$$

% Error in diameter (d):

$$(\Delta d / d) \times 100 = (0.001 / 2.000) \times 100 = 0.05\%$$

The formula for density is $\rho \propto \frac{m}{d^3}$. The percentage error is:

$$\% \text{Error}(\rho) = \% \text{Error}(m) + 3 \times \% \text{Error}(d)$$

$$\% \text{Error}(\rho) = 0.05\% + 3 \times (0.05\%)$$

$$\% \text{Error}(\rho) = 0.05\% + 0.15\% = 0.20\%$$

(NCERT Ed. 2025-26, 11th, Page No. 6)

38. (4)

$$\theta = \theta_2 - \theta_1 = 78.3 - 40.6 = 37.7^\circ \text{ C.}$$

$$\Delta \theta = \pm(\Delta \theta_1 + \Delta \theta_2)$$

$$= \pm(0.2 + 0.3) = \pm 0.5^\circ \text{ C}$$

(NCERT Ed. 2025-26, 11th, Page No. 6)

39. (4)

$$y^2 = 3x^3$$

Let $Y = y^2, X = x^3$

$Y = 3X$ is a straight line passing to origin.

$$y^2 = 3x^3$$

Let $x^3 = z$

$$y^2 = 3z \text{ represent parabola opens to the right.}$$

40. (2)

For $a \times 10^b$ and $a \leq 5$ the order of magnitude is b .

(NCERT Ed. 2025-26, 11th, Page No. 4)

41. (2)

Least count of the screw gauge

$$\text{Least count} = \frac{1 \text{ mm}}{100 \text{ divisions}} = 0.01 \text{ mm}$$

zero error = +0.05 mm

$$\text{Reading} = 4 \times 1 \text{ mm} + 60 \times 0.01 \text{ mm} - 0.05 \text{ mm}$$

$$= 4.55 \text{ mm}$$

(NCERT Ed. 2025-26, 11th, Page No. 11)

42. (3)

In 0.0021020, the underlined zeroes only are significant as this number is less than 1.

(NCERT Ed. 2025-26, 11th, Page No. 4)

43. (2)

$$P = AB + BC + CA = 3 \text{ cm} + 4 \text{ cm} + 5 \text{ cm} = 12 \text{ cm}$$

44. (1)

Only formula A and B are correct.

45. (2)

The argument W / Y must be dimensionless.

$$[W / Y] = [ML^2T^{-1}] / [ML^2T^{-1}] = [M^0L^0T^0].$$

Since the argument is dimensionless, the operation is valid.

$\sin(V + W)$: V and W cannot be added because their dimensions $[ML^{-1}T^{-3}]$ and $[ML^2T^{-1}]$ are different.

Since $[VY] \neq [Z]$, the subtraction $VY - Z$ violates the principle of dimensional homogeneity.

(NCERT Ed. 2025-26, 11th, Page No. 8)

46. (3)

- Molecules in 16 g of $O_2 = \frac{16}{32} \times N_A = 0.5N_A$
- Molecules in 1 mole of $N_2 = 1 \times N_A = N_A$
- Molecules in 5 g of $H_2 = \frac{5}{2} \times N_A = 2.5N_A$
- Molecules in 11.2 L of CO_2 at STP

$$= \frac{11.2}{22.4} \times N_A = 0.5N_A$$

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

47. (3)

$$+3 + 2x + 4(0) + 0 + 2(-1) = 0$$

$$2x = -1$$

$$x = -\frac{1}{2}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 08)

48. (2)

Since, mass of reactant (42.3 g) is equal to the mass of product (42.3 g) formed and there is no net change in mass during the process. Hence, 'Law of Conservation of Mass' is verified.

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

49. (2)

0.0830 has three significant figures.

(NCERT Ed. 2025-26 12th Part-I Page No. 08)

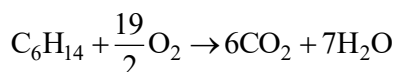
50. (2)

The oxidation state of Mg increases from 0 to +2.

An increase in oxidation state signifies oxidation.

(NCERT Ed. 2025-26 11th Part-II Page No. 240)

51. (1)



$$\text{Moles of } C_6H_{14} = \frac{129}{86} = 1.5$$

1 mole of C_6H_{14} reacts with 9.5 moles of O_2 .

1.5 mole of C_6H_{14} reacts with = 14.25 moles of O_2

Mass of $O_2 = 14.25 \times 32 = 456 \text{ g}$

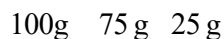
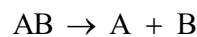
(NCERT Ed. 2025-26 11th Part-I Page No. 20)

52. (4)

Since Mg^{2+} has the highest standard reduction potential, it is the strongest oxidizing agent.

(NCERT Ed. 2025-26 12th Part-I Page No. 08)

53. (2)

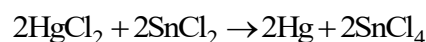


25 g of B is obtained by 100 g of AB.

Thus, 20 g of B will be obtained by $\frac{100 \times 20}{25} = 80 \text{ g}$

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

54. (3)



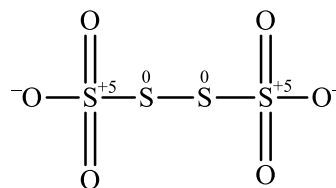
(NCERT Ed. 2025-26 11th Part-II Page No. 240)

55. (4)

%w/w, mole fraction and molality are temperature independent concentration term as they are not affected by any change in volume.

(NCERT Ed. 2025-26 11th Part-I Page No. 23)

56. (3)



Average oxidation state of S =

$$\frac{+5 + 0 + 0 + 5}{4} = \frac{10}{4} = +2.5$$

(NCERT Ed. 2025-26 11th Part-II Page No. 245)

57. (4)

Al goes from 0 to +3, so it is oxidized and is the reductant.

Fe goes from $+\frac{8}{3}$ to 0, so it is reduced and is the oxidant.

Each Al atom loses 3 electrons ($0 \rightarrow +3$).

There are 8 Al atoms, so total electrons lost by Al is $8 \times 3 = 24$ electrons.

Each Fe atom gains $\frac{8}{3}$ electrons ($+\frac{8}{3} \rightarrow 0$).

There are $3 \times 3 = 9$ Fe atoms in $3Fe_3O_4$, so total electrons gained by Fe is $9 \times \frac{8}{3} = 24$ electrons.

The number of electrons transferred from reductant to oxidant is 24.

(NCERT Ed. 2025-26 11th Part-II Page No. 240)

58. (1)

- When H is bonded to metals in binary compounds, its oxidation state number is -1.
- In elemental form, each atom bears an oxidation number of zero.

(NCERT Ed. 2025-26 11th Part-II Page No. 237)

59. (3)

56 g of CaO produces 111 g of CaCl₂

14 g of CaO will produce 27.75 g of CaCl₂

$$\text{Percentage yield} = \frac{13.875}{27.75} \times 100 = 50\%$$

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

60. (1)

$$\% \text{ of nitrogen} = \frac{\text{mass of nitrogen}}{\text{molecular weight}} \times 100$$

$$42 = \frac{14 \times x}{200} \times 100$$

$$x = \frac{42 \times 200}{14 \times 100} = 6$$

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

61. (2)

Oxidation state of C in C₆H₁₂O₆ and HCHO is zero.

(NCERT Ed. 2025-26 11th Part-II Page No. 240)

62. (1)

% abundance of ³⁵Cl = x

% abundance of ³⁷Cl = (100 - x)

$$\text{Hence, } 35.5 = \frac{35 \times x + 37 \times (100 - x)}{100}$$

$$3550 = 35x + 3700 - 37x$$

$$2x = 150$$

$$x = 75$$

Hence,

% abundance of ³⁵Cl = x = 75%

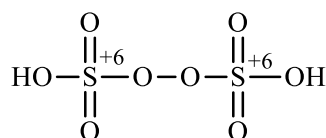
% abundance of ³⁷Cl = (100 - x) = 100 - 75 = 25%

Ratio of ³⁷Cl to ³⁵Cl = 1:3

(NCERT Ed. 2025-26 11th Part-I Page No. 16)

63. (1)

H₂S₂O₈:



(NCERT Ed. 2025-26 11th Part-II Page No. 240)

64. (3)

Mass of one drop = volume × density

$$= 9 \times 1 = 9 \text{ g}$$

$$\text{No. of moles} = \frac{9}{18} = 0.5$$

$$\text{Molecules} = 6.022 \times 10^{23} \times 0.5 = 3.011 \times 10^{23}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

65. (3)

Al₂O₃ is not a mixed oxide.

(NCERT Ed. 2025-26 12th Part-I Page No. 08)

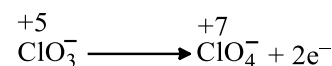
66. (4)

$$\begin{aligned} \text{Electrons in 21 g of } \text{N}_3^- \text{ ion} &= \frac{21}{42} \times N_A \times 22 \\ &= 11 N_A \end{aligned}$$

$$\text{Protons in 21 g of } \text{N}_3^- \text{ ion} = \frac{21}{42} \times N_A \times 21 = 10.5 N_A$$

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

67. (2)



$$\text{Equivalent weight} = \frac{M}{2}$$

(NCERT Ed. 2025-26 11th Part-II Page No. 240)

68. (3)

Volume of gas = 1 m³ = 10³ L

$$\text{Moles of gas} = \frac{10^3}{22.4} = 44.64$$

(NCERT Ed. 2025-26 12th Part-I Page No. 08)

69. (4)

$$M_{\text{avg}} = 2 \times VD = 2 \times 34.5 = 69 \text{ g/mol}$$

Let x be the mole fraction of NO₂.

Then (1 - x) is the mole fraction of N₂O₄.

$$M_{\text{avg}} = x \times M_{\text{NO}_2} + (1 - x) \times M_{\text{N}_2\text{O}_4}$$

$$69 = x \times 46 + (1 - x) \times 92$$

$$x = 0.5$$

Let m_{NO₂} be the mass of NO₂ and m_{N₂O₄} be the mass of N₂O₄.

$$m_{\text{NO}_2} + m_{\text{N}_2\text{O}_4} = 100 \text{ g}$$

$$\text{Moles of NO}_2 = \frac{m_{\text{NO}_2}}{46}$$

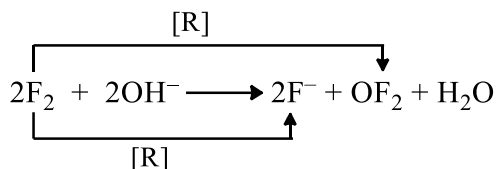
$$\text{Moles of N}_2\text{O}_4 = \frac{m_{\text{N}_2\text{O}_4}}{92}$$

$$x = \frac{\text{Moles of NO}_2}{\text{Total moles}} = \frac{\frac{m_{\text{NO}_2}}{46}}{\frac{m_{\text{NO}_2}}{46} + \frac{100 - m_{\text{NO}_2}}{92}} = 0.5$$

$$m_{\text{NO}_2} = 33.33 \text{ g}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

70. (3)



Here only reduction of F_2 is taking place
(NCERT Ed. 2025-26 11th Part-II Page No. 240)

71. (1)

Multiple	Prefix
10^{-21}	zepto
10^{-12}	pico
10^{12}	tera
10^{21}	zeta

(NCERT Ed. 2025-26 11th Part-I Page No. 09)

72. (2)

In peroxides, the oxidation state of oxygen is -1 .

- Oxidation state of O in $\text{ClO}_2^- = -2$
- Oxidation state of O in $\text{H}_2\text{O}_2 = -1$
- Oxidation state of O in $\text{KO}_2 = -\frac{1}{2}$
- Oxidation state of O in $\text{RbO}_2 = -\frac{1}{2}$

(NCERT Ed. 2025-26 11th Part-II Page No. 237)

73. (3)

- Mass = $\rho \times V = 1 \times 9 = 9 \text{ g}$
Moles = $\frac{9}{18} = 0.5$
Number of neutrons = $0.5 \times N_A \times 8$
 $= 24.088 \times 10^{23}$
- Volume of CO = $\frac{7}{28} \times 22.4 = 5.6 \text{ L} = 5600 \text{ mL}$

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

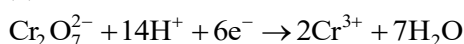
74. (3)

Number of moles left = Total moles – removed moles

$$= \frac{280}{28} - \frac{6.022 \times 10^{22}}{6.022 \times 10^{23}} = 10 - 0.1 = 9.9$$

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

75. (1)



$$\text{Equivalent weight} = \frac{M}{6}$$

(NCERT Ed. 2025-26 11th Part-II Page No. 247)

76. (1)

Element	Percentage	Atomic mass	Relative number of atoms	Simplest ratio
C	96	12	8	2
H	4	1	4	1

Empirical formula = C_2H

(NCERT Ed. 2025-26 11th Part-I Page No. 00)

77. (3)

“Equal volumes of all gases at the same temperature and pressure should contain equal number of molecules.” This statement corresponds to Avogadro’s law.

(NCERT Ed. 2025-26 11th Part-I Page No. 12)

78. (2)

Acid base neutralization reactions are non-redox reactions as oxidation number of each element remain same.

- $\overset{0}{\text{Mg}}(\text{s}) + 2\overset{+1}{\text{H}}\overset{-2}{\text{O}}(\text{l}) \rightarrow 2\overset{+2}{\text{Mg}}\overset{-2}{\text{O}}\overset{+1}{\text{H}}(\text{s}) + \overset{0}{\text{H}_2}(\text{g})$
- $\overset{+1}{\text{H}}\overset{-1}{\text{Cl}}(\text{g}) + \text{Na}\overset{+1}{\text{O}}\overset{-2}{\text{H}}(\text{s}) \rightarrow \text{Na}\overset{+1}{\text{Cl}}(\text{s}) + \overset{+1}{\text{H}}\overset{-2}{\text{O}}(\text{l})$
- $\overset{+4}{\text{Ti}}\overset{-1}{\text{Cl}_4}(\text{l}) + 2\overset{0}{\text{Mg}}(\text{s}) \rightarrow \overset{0}{\text{Ti}}(\text{s}) + 2\overset{+2}{\text{Mg}}\overset{-1}{\text{Cl}_2}(\text{s})$
- $\overset{0}{3\text{Mg}}(\text{s}) + \overset{0}{\text{N}_2}(\text{g}) \rightarrow \overset{+2}{\text{Mg}_3}\overset{-3}{\text{N}_2}(\text{s})$

(NCERT Ed. 2025-26 11th Part-II Page No. 242)

79. (1)

In Fe, oxidation state changes from $+2$ to $+3$.

In C, oxidation state changes from $+3$ to $+4$.

In Mn, oxidation state changes from $+7$ to $+2$.

(NCERT Ed. 2025-26 11th Part-I Page No. 242)

80. (4)

Given that 0.8 mol of chlorine atoms are present in 20 g of MCl_3 .

$$\text{Mass per 1 mol of chlorine atoms} = \frac{20}{0.8} = 25 \text{ g/mol}$$

Since 1 mol of MCl_3 contains 3 mol of chlorine atoms.

Molecular mass of

$$\text{MCl}_3 = 3 \times (\text{mass per 1 mol of chlorine atoms})$$

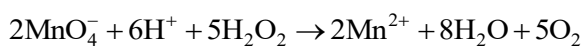
Molecular mass of

$$\text{MCl}_3 \times 25 \text{ g/mol} = 75 \text{ g/mol.}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

81. (1)

For the given reaction:



5 moles of H_2O_2 is oxidised by 2 moles of MnO_4^- .

Hence, 1 mole of H_2O_2 will be oxidised by $\frac{2}{5}$ moles of MnO_4^- .

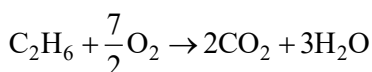
(NCERT Ed. 2025-26 11th Part-II Page No. 248)

82. (4)

HClO_4 does not undergo disproportionation reaction because Cl of HClO_4 is already in its highest oxidation state (+7).

(NCERT Ed. 2025-26 11th Part-II Page No. 244)

83. (2)



1 mol 2 mol
30 g 88g

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

84. (1)

F_2 is the strongest oxidising agent while Li is the strongest reducing agent.

(NCERT Ed. 2025-26 11th Part-II Page No. 251)

85. (2)

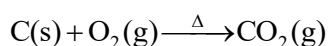
$$\text{Vapour density} = \frac{\text{molecular mass}}{2}$$

$$\frac{M_A}{M_B} = \frac{VD_A}{VD_B} = \frac{1}{2}$$

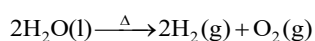
(NCERT Ed. 2025-26 11th Part-I Page No. 19)

86. (1)

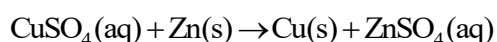
- Combination reaction:



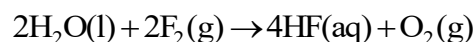
- Decomposition reaction:



- Metal displacement reaction:



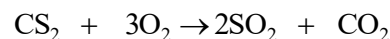
- Non-metal displacement reaction:



(NCERT Ed. 2025-26 11th Part-II Page No. 242)

87. (3)

For the reaction,



$$\text{moles: } \frac{38}{76} = 0.5 \quad \frac{64}{32} = 2$$

Here, limiting reagent is CS_2

Since, 1 moles of CS_2 gives 2 moles of SO_2 .

Thus, 0.5 mole of CS_2 will give 1 mole of SO_2 .

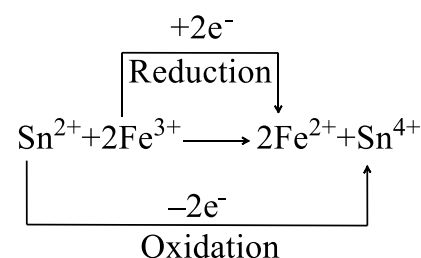
Mass of $\text{SO}_2 = 1 \times 64 = 64 \text{ g}$

$$\% \text{ yield} = \frac{\text{experimental mass}}{\text{theoretical mass}} \times 100$$

$$= \frac{16}{64} \times 100 = 25\%$$

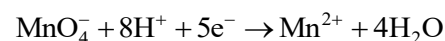
(NCERT Ed. 2025-26 11th Part-I Page No. 21)

88. (1)



(NCERT Ed. 2025-26 11th Part-II Page No. 241)

89. (1)



$$\text{Equivalent weight} = \frac{M}{5}$$

(NCERT Ed. 2025-26 11th Part-II Page No. 247)

90. (2)

- Oxidation state of Mn in $\text{KMnO}_4 = +7$
- Oxidation state of Mn in $\text{K}_2\text{MnO}_4 = +6$
- Oxidation state of Mn in $\text{MnO}_2 = +4$
- Oxidation state of Mn in $\text{Mn}_2\text{O}_3 = +3$

(NCERT Ed. 2025-26 11th Part-II Page No. 250)

91. (1)

The outer membrane of the nuclear envelope usually remains continuous with the endoplasmic reticulum and also bears ribosomes on it. The content of nucleolus is continuous with the rest of the nucleoplasm as it is not a membrane bound structure. It is a site for active ribosomal RNA synthesis. The nuclear pores are the passages through which movement of RNA and protein molecules takes place in both directions between the nucleus and the cytoplasm. The interphase nucleus has a loose and indistinct network of nucleoprotein fibres called chromatin.

(NCERT Ed. 2025-26, 11th, Page No. 100)

92. (3)

The smooth endoplasmic reticulum is the major site for synthesis of lipid. In animal cells lipid-like steroidal hormones are synthesised in SER.

(NCERT Ed. 2025-26, 11th, Page No. 95)

93. (1)

Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). Theodore Schwann concluded, based on his studies on plant tissues, that the presence of cell wall is a unique character of the plant cells. Robert Brown discovered the nucleus. Antonie Von Leeuwenhoek first saw and described a live cell.

(NCERT Ed. 2025-26, 11th, Page No. 88)

94. (2)

Several ribosomes may attach to a single mRNA and form a chain called polyribosomes or polysome. The ribosomes of a polysome translate the mRNA into proteins.

(NCERT Ed. 2025-26, 11th, Page No. 91)

95. (3)

The middle lamella is a layer mainly of calcium pectate which holds or glues the different neighbouring cells together. Therefore, dissolving this layer would remove the adhesion between cells, causing them to separate.

(NCERT Ed. 2025-26, 11th, Page No. 94)

96. (4)

Mycoplasmas, the smallest cells, are only 0.3 μm in lengths. Typical bacteria is 1-2 μm . The human red blood cells are about 7.0 μm in diameter. A typical

eukaryotic cell size is 10-20 μm . The largest isolated single cell is the egg of an ostrich.

(NCERT Ed. 2025-26, 11th, Page No. 89)

97. (2)

The eukaryotic ribosomes are 80S. Here 'S' (Svedberg's unit) stands for the sedimentation coefficient; it is indirectly a measure of density and size of the ribosome. The two subunits of 80S ribosomes are 60S and 40S while that of 70S ribosomes are 50S and 30S.

(NCERT Ed. 2025-26, 11th, Page No. 98)

98. (1)

Besides flagella, Pili and Fimbriae are also surface structures of the bacteria but do not play a role in motility.

(NCERT Ed. 2025-26, 11th, Page No. 91)

99. (2)

The cell envelope in bacteria consists of a tightly bound three layered structure. Although each layer of the envelope performs distinct function, they act together as a single protective unit. The cell envelope consists of a tightly bound three layered structure i.e., the outermost glycocalyx followed by the cell wall and then the plasma membrane. Glycocalyx differs in composition and thickness among different bacteria. It could be a loose sheath called the slime layer in some, while in others it may be thick and tough, called the capsule. The cell wall determines the shape of the cell and provides a strong structural support to prevent the bacterium from bursting or collapsing. All prokaryotes have a cell wall surrounding the cell membrane except in mycoplasma.

(NCERT Ed. 2025-26, 11th, Page No. 90)

100. (2)

Centrosome is an organelle usually containing two cylindrical structures called centrioles. They are surrounded by amorphous pericentriolar materials. Both the centrioles in a centrosome lie perpendicular to each other in which each has an organisation like the cartwheel. The central part of the proximal region of the centriole is also proteinaceous and called the hub, which is connected with tubules of the peripheral triplets by radial spokes made of protein.

(NCERT Ed. 2025-26, 11th, Page No. 99)

101. (2)

The enzymes in lysosomes are capable of digesting carbohydrates, proteins, lipids and nucleic acids. Mitochondria are the sites of aerobic respiration. Golgi apparatus is the important site of formation of glycoproteins and glycolipids. The vacuole is the membrane-bound space found in the cytoplasm. It contains water, sap, excretory product and other materials not useful for the cell.

(NCERT Ed. 2025-26, 11th, Page No. 96)

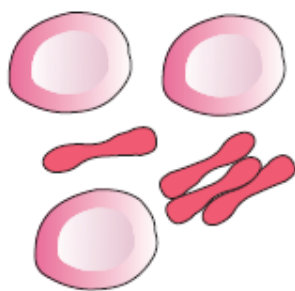
102. (4)

The inner membrane of mitochondria forms a number of infoldings called the cristae (sing.: crista) towards the matrix. The cristae increase the surface area.

(NCERT Ed. 2025-26, 11th, Page No. 97)

103. (4)

In animal cells, cell wall is not present.



Red blood cells
(round and biconcave)

(NCERT Ed. 2025-26, 11th, Page No. 93)

104. (1)

An improved model of the structure of cell membrane was proposed by Singer and Nicolson (1972) widely accepted as fluid mosaic model. According to this, the quasi-fluid nature of lipid enables lateral movement of proteins within the overall bilayer. The fluid nature of the membrane is also important from the point of view of functions like cell growth, formation of intercellular junctions, secretion, endocytosis, cell division etc.

(NCERT Ed. 2025-26, 11th, Page No. 94)

105. (2)

In some prokaryotes like cyanobacteria, there are other membranous extensions into the cytoplasm called chromatophores which contain pigments.

Plastids are found in all plant cells and in euglenoides. These are easily observed under the microscope as they are large. They bear some specific pigments, thus imparting specific colours to the plants. Based on the type of pigments plastids can be classified into chloroplasts, chromoplasts and leucoplasts. The chloroplasts contain chlorophyll and carotenoid pigments which are responsible for trapping light energy essential for photosynthesis. In the chromoplasts, fat soluble carotenoid pigments like carotene, xanthophylls and others are present. This gives the part of the plant a yellow, orange or red colour. The leucoplasts are the colourless plastids of varied shapes and sizes with stored nutrients.

(NCERT Ed. 2025-26, 11th, Page No. 97)

106. (2)

A non-living rigid structure called the cell wall forms an outer covering for the plasma membrane of fungi and plants. Cell wall not only gives shape to the cell and protects the cell from mechanical damage and infection. The cell wall of a young plant cell, the primary wall is capable of growth.

(NCERT Ed. 2025-26, 11th, Page No. 94)

107. (1)

The metacentric chromosome has middle centromere forming two equal arms of the chromosome. The sub-metacentric chromosome has centromere slightly away from the middle of the chromosome resulting into one shorter arm and one longer arm. In case of acrocentric chromosome the centromere is situated close to its end forming one extremely short and one very long arm, whereas the telocentric chromosome has a terminal centromere.

(NCERT Ed. 2025-26, 11th, Page No. 101)

108. (4)

A few ions or molecules are transported across the membrane against their concentration gradient, i.e., from lower to the higher concentration. Such a transport is an energy dependent process, in which ATP is utilised and is called active transport, e.g., Na^+/K^+ Pump.

(NCERT Ed. 2025-26, 11th, Page No. 94)

109. (2)

The endomembrane system include endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles. Since the functions of the mitochondria, chloroplast and peroxisomes are not coordinated with the above components, these are not considered as part of the endomembrane system.

(NCERT Ed. 2025-26, 11th, Page No. 95)

110. (3)

Materials to be packaged in the form of vesicles from the ER fuse with the *cis* face of the golgi apparatus and move towards the maturing face.

(NCERT Ed. 2025-26, 11th, Page No. 96)

111. (2)

Ribosomes are non-membrane bound organelles found in all cells – both eukaryotic as well as prokaryotic. Within the cell, ribosomes are found not only in the cytoplasm but also within the two organelles – chloroplasts (in plants) and mitochondria and on rough ER.

(NCERT Ed. 2025-26, 11th, Page No. 88)

112. (3)

The cell wall and middle lamellae may be traversed by plasmodesmata which connect the cytoplasm of neighbouring plant cells. Microvilli are finger-like projections in animal cells. Reserve material in prokaryotic cells are stored in the cytoplasm in the form of inclusion bodies. These are not bound by any membrane system and lie free in the cytoplasm, e.g., phosphate granules, cyanophycean granules and glycogen granules. Algae have cell wall, made of cellulose, galactans, mannans and minerals like calcium carbonate.

(NCERT Ed. 2025-26, 11th, Page No. 94)

113. (3)

A specialised differentiated form of cell membrane called mesosome is the characteristic of prokaryotes. They are essentially infoldings of cell membrane. They help in cell wall formation, DNA replication and distribution to daughter cells. They also help in respiration, secretion processes, to increase the surface area of the plasma membrane and enzymatic content. The plasmid DNA confers certain unique phenotypic characters to bacteria. One such character is resistance to antibiotics.

(NCERT Ed. 2025-26, 11th, Page No. 91)

114. (3)

A number of proteins synthesised by ribosomes on the endoplasmic reticulum are modified in the cisternae of the golgi apparatus before they are released from its *trans* face. Materials to be packaged in the form of vesicles from the ER fuse with the *cis* face of the golgi apparatus and move towards the maturing face.

(NCERT Ed. 2025-26, 11th, Page No. 96)

115. (4)

The stroma lamellae connect the thylakoids of the different grana. The electron microscopic study of a cilium or the flagellum show that they are covered with plasma membrane. Their core called the axoneme, possesses a number of microtubules running parallel to the long axis. The axoneme usually has nine doublets (i.e. 18) of radially arranged peripheral microtubules, and a pair (2) of centrally located microtubules. Such an arrangement of axonemal microtubules is referred to as the 9+2 array.

(NCERT Ed. 2025-26, 11th, Page No. 99)

116. (1)

The number of mitochondria per cell is variable depending on the physiological activity of the cells. The matrix also possesses single circular DNA molecule, a few RNA molecules, ribosomes (70S) and the components required for the synthesis of proteins. The mitochondria divide by fission. Mitochondria (sing.: mitochondrion), unless specifically stained, are not easily visible under the microscope.

(NCERT Ed. 2025-26, 11th, Page No. 97)

117. (2)

In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the cell. The cytoplasm is the main arena of cellular activities in both the plant and animal cells. Various chemical reactions occur in it to keep the cell in the 'living state'. In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called plasmids.

(NCERT Ed. 2025-26, 11th, Page No. 88)

118. (4)

Lysosomes are membrane bound vesicular structures formed by the process of packaging in the golgi apparatus. ER divides the intracellular space into two distinct compartments, i.e., luminal (inside ER) and extra luminal (cytoplasm) compartments. In eukaryotic cells, genetic material is present inside nucleus. The smooth endoplasmic reticulum is the major site for synthesis of lipid. In animal cells lipid-like steroidal hormones are synthesised in SER. RER bear ribosomes on surface.

(NCERT Ed. 2025-26, 11th, Page No. 95)

119. (3)

Biology is the study of living organisms. The detailed description of their form and appearance only brought out their diversity. It is the cell theory that emphasised the unity underlying this diversity of forms, i.e., the cellular organisation of all life forms.

(NCERT Ed. 2025-26, 11th, Page No. 87)

120. (2)

RER is frequently observed in the cells actively involved in protein synthesis and secretion. The endoplasmic reticulum bearing ribosomes on their surface is called rough endoplasmic reticulum (RER). Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.

(NCERT Ed. 2025-26, 11th, Page No. 95)

121. (3)

The vacuole is the membrane-bound space found in the cytoplasm. It contains water, sap, excretory product and other materials not useful for the cell. The vacuole is bound by a single membrane called tonoplast. In plant cells the vacuoles can occupy up to 90 per cent of the volume of the cell. In plants, the tonoplast facilitates the transport of a number of ions and other materials against concentration gradients into the vacuole. The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes (hydrolases – lipases, proteases, carbohydrases) optimally active at the acidic pH. An elaborate network of filamentous proteinaceous structures consisting of microtubules, microfilaments and intermediate

filaments present in the cytoplasm is collectively referred to as the cytoskeleton.

(NCERT Ed. 2025-26, 11th, Page No. 96)

122. (4)

Cilia (sing.: cilium) and flagella (sing.: flagellum) are hair-like outgrowths of the cell membrane. Both the cilium and flagellum emerge from centriole-like structure called the basal bodies. Animal cells contain another non-membrane bound organelle called centrosome which helps in cell division. Animal cells have centrioles which are absent in almost all plant cells. The centrioles form the basal body of cilia or flagella, and spindle fibres that give rise to spindle apparatus during cell division in animal cells.

(NCERT Ed. 2025-26, 11th, Page No. 99)

123. (2)

Electron microscopy has revealed that the nuclear envelope, which consists of two parallel membranes with a space between (10 to 50 nm) called the perinuclear space, forms a barrier between the materials present inside the nucleus and that of the cytoplasm. Typically, mitochondrion is sausage-shaped or cylindrical having a diameter of 0.2-1.0 μ m (average 0.5 μ m) and length 1.0-4.1 μ m. Golgi bodies consist of many flat, disc-shaped sacs or cisternae of 0.5 μ m to 1.0 μ m diameter. In prokaryotes, ribosomes are associated with the plasma membrane of the cell. They are about 15 nm by 20 nm in size.

(NCERT Ed. 2025-26, 11th, Page No. 100)

124. (4)

In plants, the tonoplast facilitates the transport of a number of ions and other materials against concentration gradients into the vacuole. In *Amoeba*, the contractile vacuole is important for osmoregulation and excretion.

(NCERT Ed. 2025-26, 11th, Page No. 96)

125. (1)

Many membrane bound minute vesicles called microbodies that contain various enzymes, are present in both plant and animal cells.

(NCERT Ed. 2025-26, 11th, Page No. 102)

126. (1)

Plastids are found in all plant cells and in euglenoides. The leucoplasts are the colourless plastids of varied shapes and sizes with stored nutrients: amyloplasts store carbohydrates (starch) e.g., potato; elaioplasts store oils and fats whereas the aleuroplasts store proteins. Like mitochondria, the chloroplasts are also double membrane bound. The plasma membrane in all organisms is mainly made up of phospholipids, and also contain proteins, cholesterol etc.

(NCERT Ed. 2025-26, 11th, Page No. 97)

127. (4)

Sometimes a few chromosomes have non-staining secondary constrictions at a constant location. This gives the appearance of a small fragment called the satellite. Chromatin contains DNA and some basic proteins called histones, some non-histone proteins and also RNA. A single human cell has approximately two metre long thread of DNA distributed among its forty six (twenty three pairs) chromosomes. Every chromosome (visible only in dividing cells) essentially has a primary constriction or the centromere on the sides of which disc shaped structures called kinetochores are present. Centromere holds two chromatids of a chromosome.

(NCERT Ed. 2025-26, 11th, Page No. 101)

128. (2)

The genetic material in prokaryotes is basically naked, not enveloped by a nuclear membrane. Chromatin is present in eukaryotic cells.

(NCERT Ed. 2025-26, 11th, Page No. 90)

129. (3)

All eukaryotic cells are not identical. Plant and animal cells are different as the former possess cell walls, plastids and a large central vacuole which are absent in animal cells. On the other hand, animal cells have centrioles which are absent in almost all plant cells. In addition, eukaryotic cells have a variety of complex locomotory and cytoskeletal structures.

(NCERT Ed. 2025-26, 11th, Page No. 91)

130. (3)

Theodore Schwann (1839), a German Zoologist, studied different types of animal cells and reported that cells had a thin outer layer which is today known as the 'plasma membrane'.

(NCERT Ed. 2025-26, 11th, Page No. 88)

131. (1)

Depending on the ease of extraction, membrane proteins can be classified as integral and peripheral. Peripheral proteins lie on the surface of membrane while the integral proteins are partially or totally buried in the membrane. The major lipids are phospholipids that are arranged in a bilayer. Also, the lipids are arranged within the membrane with the polar head towards the outer sides and the hydrophobic tails towards the inner part. This ensures that the non-polar tail of saturated hydrocarbons is protected from the aqueous environment.

(NCERT Ed. 2025-26, 11th, Page No. 93)

132. (4)

Unicellular organisms are capable of (i) independent existence and (ii) performing the essential functions of life. All cells arise from pre-existing cells, the unicellular cells also divide to produce new cells.

(NCERT Ed. 2025-26, 11th, Page No. 87)

133. (3)

Like mitochondria, the chloroplasts are also double membrane bound. Of the two, the inner chloroplast membrane is relatively less permeable. Plastids are found in all plant cells and in euglenoides. These are easily observed under the microscope as they are large. Chlorophyll pigments are present in the thylakoids. The ribosomes of the chloroplasts are smaller (70S) than the cytoplasmic ribosomes (80S).

(NCERT Ed. 2025-26, 11th, Page No. 98)

134. (2)

The cytoplasm of a eukaryotic cell (like a plant cell) has 80S ribosomes. Mitochondria and chloroplasts contain their own 70S ribosomes. The toxin inhibits 40S ribosomes (one of the sub units of 80S ribosome), so the synthesis of proteins in the cytoplasm will stop. However, the 70S ribosomes within the mitochondria will remain functional and can continue to synthesize proteins.

(NCERT Ed. 2025-26, 11th, Page No. 98)

135. (3)

Bacteria can be classified into two groups on the basis of the differences in the cell envelopes and the manner in which they respond to the staining procedure developed by Gram viz., those that take up the gram stain are Gram positive and the others that do not are called Gram negative bacteria.

(NCERT Ed. 2025-26, 11th, Page No. 90)

136. (2)

- Each organ system is formed by multiple organs, each composed of different tissue types.
- These organs are linked physically (by structural connections such as ducts, blood vessels, or nerves) and/or chemically (through hormones, enzymes, neurotransmitters, etc.).

Example:

- Digestive system: Mouth, stomach, intestines, liver, and pancreas work together to digest food and absorb nutrients.

(NCERT Ed. 2025-26, 11th, Page No. 79)

137. (3)

Structure of dense regular connective tissue:

- Fibres: Contains thick collagen fibres arranged in parallel bundles, which ensures strength in a single direction.
- Cells: Fibroblasts are arranged in rows between these bundles, actively secreting collagen.
- Matrix: Relatively less ground substance due to the dense packing of fibres.

(NCERT Ed. 2020-21, 11th, Page No. 103)

138. (1)

List-I	List-II
Tendons	Attach muscles to bones
Fibroblasts	Fibres secreting cells
Hyaline cartilage	Tip of nose
Ligaments	Attach bones to bones

(NCERT Ed. 2020-21, 11th, Page No. 103)

139. (2)

- Chondrocytes are the cells of cartilage, and they are located in small spaces called lacunae within the cartilage's pliable (flexible) and avascular matrix.
- Bone is a hard, rigid tissue due to its matrix being rich in calcium salts (Ca^{2+}), not low calcium. It is also rich in collagen fibers, which provide tensile strength, but not elastin, which provides elasticity.
- Osteocytes are the mature cells of bone tissue. They are located in lacunae and play a vital role in maintaining the bone matrix by regulating its mineral and collagen composition.
- Cartilage is found in several locations in the adult body. For example, hyaline cartilage is found in synovial joints, elastic cartilage is found in the pinna (external ear), and fibrocartilage is found in the intervertebral discs, where it acts as a shock absorber.

(NCERT Ed. 2020-21, 11th, Page No. 104)

140. (1)

- The cellular organization in multicellular organisms into tissues, organs, and organ systems provides a significant advantage. This level of organization allows for greater complexity and specialization, which improves the organism's ability to adapt and survive in various environments.
- The formation of tissues enables a division of labor. For example, muscle tissue specializes in contraction for movement, while nervous tissue specializes in transmitting signals.

(NCERT Ed. 2025-26, 11th, Page No. 79)

141. (2)

- Macrophages are phagocytic cells. They are a crucial part of the immune system, engulfing and digesting cellular debris, foreign substances, microbes, and cancer cells.
- Mast cells are responsible for secreting histamine. Histamine is a chemical mediator involved in allergic reactions and inflammation. When a mast cell is triggered, it releases histamine, which causes blood vessels to dilate and become more permeable, allowing immune cells and proteins to reach the site of inflammation.

(NCERT Ed. 2020-21, 11th, Page No. 103)

142. (4)

Haversian canals, also known as central canals, are a unique feature of mature mammalian bone (specifically, compact bone). They are tiny channels that run longitudinally through the bone tissue, containing blood vessels and nerves. This organized structure is called an osteon and is a key feature that distinguishes bone from cartilage.

(NCERT Ed. 2020-21, 11th, Page No. 104)

143. (2)

Areolar tissue is a type of loose connective tissue. Its primary function is to serve as a soft, flexible packing material between organs and tissues. It acts as a supportive framework for epithelial tissue, holding it in place while also allowing for some movement and flexibility. It also provides a passageway for blood vessels and nerves.

(NCERT Ed. 2020-21, 11th, Page No. 103)

144. (2)

- Tendons connect muscles to bones and are subject to forces primarily in one direction. Their structure of densely packed, parallel collagen fibers provides great tensile strength in that single direction.
- Ligaments, which connect bone to bone, are actually made of dense regular connective tissue, with their fibers arranged in parallel, similar to tendons. This arrangement provides them with the strength to withstand the unidirectional stress of stabilizing joints.
- The dense irregular connective tissue, on the other hand, has collagen fibers oriented randomly and is found in locations like the dermis of the skin, where strength is needed in multiple directions.

(NCERT Ed. 2020-21, 11th, Page No. 103)

145. (2)

Fibroblasts are the most common cells in connective tissue and play a critical role in wound healing. They are characterized by their elongated, spindle shape and their primary function is to synthesize and secrete the components of the extracellular matrix. These components include:

- Protein fibers, such as collagen and elastin, which form a meshwork that provides structural support.
- Ground substance, a gel-like substance composed of glycoproteins and polysaccharides that fills the spaces between the cells and fibers.

During wound healing, fibroblasts migrate to the injury site and become highly active, producing the collagen-rich tissue that will form the scar.

(NCERT Ed. 2020-21, 11th, Page No. 103)

146. (3)

- Dendrites are the receptive or input regions of a neuron. They receive nerve impulses from other neurons and transmit them toward the cell body.
- An axon is the long, slender projection of a nerve cell that conducts nerve impulses away from the cell body toward other neurons or effector cells (like muscle cells).
- Mature mammalian red blood cells (erythrocytes) are enucleated (they lack a nucleus) whereas white blood cells (leukocytes) have nuclei.
- Skeletal muscle tissue is composed of numerous long, cylindrical fibers arranged in a parallel fashion, which allows for coordinated contraction and movement.

(NCERT Ed. 2025-26, 11th, Page No. 82)

147. (3)

- In adipose tissue, the fat vacuole occupies most of the cytoplasm, pushing the nucleus to the periphery (marginalised appearance).
- Compound epithelium is made of more than one layer (multi-layered) of cells and thus has a limited role in secretion and absorption.
- Pseudostratified epithelium appears multilayered due to nuclei at different levels, but all cells touch the basement membrane. It is a type of simple epithelium, not stratified, because there is only one true layer of cells.
- In connective tissue, the nature of the matrix (solid in bone, pliable in cartilage, fluid in blood) determines its rigidity, flexibility, or fluidity.

(NCERT Ed. 2020-21, 11th, Page No. 101)

148. (3)

The stomach is lined by simple columnar epithelium with mucus-secreting cells, not compound epithelium. Compound (stratified) epithelium is designed mainly for protection in areas subjected to mechanical and chemical stress (e.g., skin, buccal cavity, pharynx).

The primary role of compound epithelium is protection, not active absorption, secretion, or excretion. Simple epithelium types (squamous, cuboidal, columnar) are specialised for those functions.

(NCERT Ed. 2020-21, 11th, Page No. 102)

149. (3)

The bones in the limbs, such as the femur (thigh bone) and humerus (upper arm bone), are classified as long bones. The primary functions of long bones are to support body weight and act as levers for muscle action to produce movement. The bone marrow in some bones is the site of production of blood cells. Bones, particularly flat bones like the skull, ribs, and pelvis, provide protection for vital, softer tissues and organs such as the brain, heart, and lungs.

(NCERT Ed. 2020-21, 11th, Page No. 104)

150. (2)

Intercalated discs in cardiac muscle facilitate rapid and coordinated transmission of electrical impulses between cells, enabling synchronized heart contractions. Destruction of these discs disrupts electrical coupling, leading to irregular heartbeats and impaired cardiac function. This effect is specific to cardiac muscle and does not affect skeletal muscle striations, smooth muscle control, or bone strength.

(NCERT Ed. 2020-21, 11th, Page No. 105)

151. (2)

Both neurons and muscle fibers are "excitable" cells. They can be stimulated to generate and propagate an electrical signal, which is a key characteristic of these tissues.

- The neuron is the structural and functional unit of the nervous system. It is specialized to transmit information via electrical and chemical signals.
- Neuroglia, or glial cells, are very abundant and make up a significant portion of the nervous tissue. In many parts of the nervous system, they outnumber neurons and account for more than 50% of the tissue's volume.
- Arrival of the disturbance at the neuron's endings, or output zone, triggers events that may cause stimulation or inhibition of adjacent neurons and other cells.
- The neural tissue, comprising the brain, spinal cord, and nerves, is responsible for detecting changes in the internal and external environment, processing that information, and initiating a response, often through the motor system (e.g., muscles).

(NCERT Ed. 2020-21, 11th, Page No. 105)

152. (4)

- The outermost layer of the skin is composed of stratified squamous epithelium, which provides protection.
- The bronchioles in the respiratory system are lined with ciliated epithelium to move mucus and trapped particles out of the airways.
- Glandular epithelium is specialized for secretion and forms the primary tissue of glands like the salivary glands.
- The lining of body cavities (e.g., peritoneal, pleural) is composed of simple squamous epithelium called the mesothelium, which allows for smooth movement and fluid exchange. Columnar epithelium is found in organs like the stomach and intestines, where its function is absorption and secretion

(NCERT Ed. 2020-21, 11th, Page No. 101)

153. (3)

The tissue discussed in the question is brush bordered cuboidal epithelium. The tubular structure where it is found is PCT of kidney. There these microvilli increase surface area for reabsorbing water, ions, and nutrients from the filtrate back into the blood.

(NCERT Ed. 2020-21, 11th, Page No. 101)

154. (3)

The diameter of a blood vessel is controlled by the layer of smooth muscle in its walls. Smooth muscles are involuntary, meaning they are not under conscious control, and their contraction

(vasoconstriction) and relaxation (vasodilation) regulate blood pressure and blood flow.

(NCERT Ed. 2020-21, 11th, Page No. 105)

155. (2)

Keratin is a fibrous, structural protein that provides strength and waterproofing to epithelial cells. It's particularly abundant in the keratinized stratified squamous epithelium that makes up the epidermis, the outermost layer of the skin. The presence of keratin gives skin its tough, protective, and water-resistant properties. Without keratin, the skin's primary function as a barrier against pathogens, physical abrasion, and water loss would be severely compromised.

(NCERT Ed. 2020-21, 11th, Page No. 102)

156. (4)

- Neurons are specialised for impulse conduction and are arranged in networks rather than layers like epithelial cells. Their structural arrangement is individual and interconnected via synapses, not in compact layers.
- Smooth muscle in blood vessels contracts independently under autonomic and local control; it does not contract in unison like cardiac muscle. Cardiac muscle is unique in its syncytial contraction due to intercalated discs and gap junctions, ensuring simultaneous action.
- Epithelial tissues are avascular (lack blood vessels). They receive nutrition via diffusion from underlying connective tissues through the basement membrane.
- Connective tissue is the most abundant in the body, providing support, binding, protection, storage, and transport. Examples: blood, bone, cartilage, adipose, areolar tissue.

(NCERT Ed. 2020-21, 11th, Page No. 102)

157. (4)

Gap junctions are found in:

1. Cardiac muscle – Present in intercalated discs, allowing rapid spread of action potentials for synchronous contraction.
 2. Single-unit smooth muscle – Found in the walls of visceral organs (e.g., intestines, uterus, small arteries) for coordinated contraction.
 3. Certain epithelial tissues – Present to coordinate ion and fluid transport between adjacent epithelial cells.
- Differentiated skeletal muscle fibres do not have gap junctions. Skeletal muscle fibres are multinucleated and function independently; each fibre is individually innervated by a motor neuron at the neuromuscular junction. Coordination in skeletal muscle is achieved by the nervous system, not by direct cytoplasmic connections between fibres.

(NCERT Ed. 2020-21, 11th, Page No. 102)

158. (2)

- All three types of connective tissue (Areolar, dense irregular, and dense regular tissues) contain collagen, the most abundant protein in the body, which provides tensile strength.
- Fibroblasts are the primary cells in these tissues, responsible for producing and secreting the fibers and ground substance.
- Areolar, dense irregular, and dense regular tissues are classified as types of connective tissue.
- Only dense regular connective tissue forms tendons.
- Similar to tendons, ligaments are primarily made of dense regular connective tissue.
- Areolar tissue is highly vascular, but dense regular and dense irregular connective tissues are less vascular.
- All connective tissues, including areolar, dense irregular, and dense regular, are derived from the mesoderm, one of the three primary germ layers in a developing embryo.

(NCERT Ed. 2020-21, 11th, Page No. 103)

159. (2)

List-I	List-II
Heart	Organ that consists of all four types of tissues
Cockroach	Represents invertebrates
Frog	Represents vertebrates
Electrical disturbance generated in a neuron	Travels along plasma membrane

(NCERT Ed. 2020-21, 11th, Page No. 106)

160. (4)

- Goblet cells are unicellular exocrine glands present in the lining of the respiratory and intestinal tracts. They secrete mucus for lubrication and protection.
- The germinal epithelium covering the ovary is simple cuboidal epithelium.
- Areolar tissue is a loose connective tissue found beneath epithelia. It binds skin to underlying muscles and fills spaces between organs.
- Eosinophils have a bilobed nucleus (two lobes connected by a thin chromatin strand), not multilobed. Multilobed nuclei are characteristic of neutrophils (3–5 lobes).

(NCERT Ed. 2020-21, 11th, Page No. 102)

161. (2)

- The liver has a remarkable capacity for regeneration. If a portion of the liver is damaged or removed, the remaining hepatocytes can re-

enter the cell cycle and divide to replace the lost tissue.

- In adults, neurons have lost the ability to divide, therefore, if damaged or die they are least likely to be replaced.
- Skin has a very high regenerative capacity. The basal layer of the epidermis contains stem cells that are continuously dividing to produce new cells that move up, differentiate, and replace the older cells that are sloughed off from the surface.
- Mature osteocytes themselves are trapped in the bone matrix and do not divide but bone tissue as a whole has a good capacity for repair and remodeling. This is due to the presence of other cells, such as osteoblasts, which can divide and produce new bone matrix to heal fractures and replace damaged areas

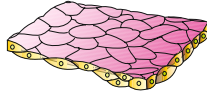
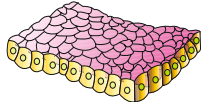
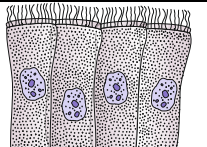
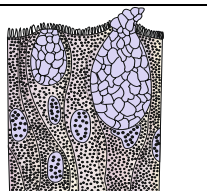
(NCERT Ed. 2020-21, 11th, Page No. 105)

162. (1)

- Adipose tissue is the body's long-term energy store. Adipocytes synthesize triglycerides from fatty acids and glucose (lipogenesis) and store them as a large cytoplasmic lipid droplet, to be mobilized later (lipolysis).
- Chondrocytes (cartilage) secrete collagen and proteoglycans (not calcium salts); osteocytes/osteoblasts (bone) are associated with mineralized matrix containing calcium salts. Adipose tissue consists mainly of adipocytes within a loose connective tissue stroma (capillaries, fibroblasts, macrophages), not a semi-rigid, mineralized matrix.

(NCERT Ed. 2020-21, 11th, Page No. 103)

163. (4)

List-I	List-II
	Pavement epithelium
	Cuboidal epithelium
	Ciliated columnar epithelium
	Unicellular gland

(NCERT Ed. 2020-21, 11th, Page No. 101)

164. (3)

- Gap junctions allow the rapid transfer of ions and small molecules, but they also facilitate the passage of large molecules to a certain extent, and are crucial for cell to cell communication.
- Specialized junctions, including tight junctions, adhering junctions, and gap junctions, are essential in almost all animal tissues to provide both structural links (e.g., holding cells together) and functional links (e.g., allowing communication and transport).
- Tight junctions create a seal between adjacent cells, preventing the leakage of fluids and other substances across a tissue. This is particularly important in tissues that line cavities, such as the stomach and intestines.
- Adhering junctions (also known as desmosomes and adherens junctions) provide strong structural support and cementing neighbouring cells together.
- Gap junctions form channels that directly connect the cytoplasm of adjacent cells, allowing for the passage of ions and molecules. This enables cells to communicate with each other and coordinate their activities, for example, in the heart muscle.

(NCERT Ed. 2020-21, 11th, Page No. 102)

165. (1)

Bone has a calcified, collagen-rich matrix arranged in concentric lamellae around Haversian canals (osteons), enabling great strength.

Cartilage has a solid but pliable matrix that is not arranged in concentric rings; chondrocytes lie in lacunae scattered within the homogeneous matrix without a Haversian system.

(NCERT Ed. 2020-21, 11th, Page No. 104)

166. (2)

Blood is a fluid connective tissue with plasma as the liquid matrix. Areolar tissue is a loose connective tissue with a soft, gel-like matrix. Cartilage has solid but pliable matrix rich in chondroitin sulfate and collagen fibres. Bone has matrix calcified with calcium phosphate salts and collagen fibres.

(NCERT Ed. 2020-21, 11th, Page No. 104)

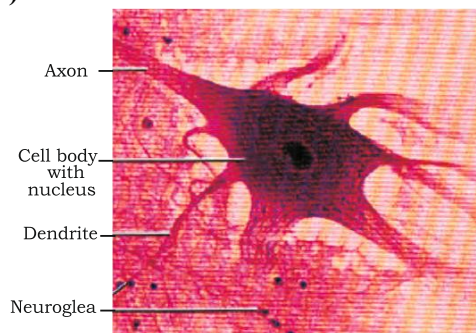
167. (4)

- The progression from tissue level (e.g., cnidarians) → organ level (e.g., flatworms) → organ-system level (e.g., annelids, vertebrates) represents a clear directional pattern in evolution.
- This pattern shows how multicellular organisms have evolved more complex and specialized structures over time to perform functions more efficiently.
- This is not just about physical form (morphological) or internal structure

(anatomical), but about a historical, evolutionary progression observed across the animal kingdom.

(NCERT Ed. 2025-26, 11th, Page No. 80)

168. (1)



A – Axon, B - Neuroglia

(NCERT Ed. 2020-21, 11th, Page No. 105)

169. (3)

- In adults, the vertebral column is primarily bony, with cartilage found only in intervertebral discs (fibrocartilage) and at certain joint surfaces. Cartilaginous vertebral columns are found in embryos, not adults.
- In cartilage, chondrocytes are located within small spaces called lacunae. The matrix is solid but pliable due to the presence of proteoglycans and collagen fibres, allowing flexibility while maintaining shape.
- Bone contains osteocytes within lacunae, surrounded by a hard matrix made of calcium phosphate salts and collagen fibres for rigidity and tensile strength.
- Red bone marrow is the site of formation of blood cells (RBCs, WBCs, and platelets) in adults, especially in flat bones and the ends of long bones.

(NCERT Ed. 2020-21, 11th, Page No. 104)

170. (2)

The erector pili muscles, which cause hair to stand on end, are a classic example of smooth muscle, not striated muscle. They are composed of uninucleate, non-striated fibres. The action of these muscles is entirely involuntary, controlled by the autonomic nervous system in response to cold or fear. It is not under voluntary control.

The wall of the bladder (detrusor muscle) is made of smooth muscle (uninucleate, non-striated) and its contraction is involuntary.

The biceps is a skeletal muscle (multinucleate, striated) and its contraction to bend the arm is under voluntary control.

The wall of blood vessels (middle layer) contains smooth muscle (uninucleate, fusiform/spindle-shaped) that contracts or relaxes involuntarily to regulate blood pressure and flow.

(NCERT Ed. 2020-21, 11th, Page No. 105)

171. (1)

List-I	List-II
<i>Hydra</i>	Simplest example of tissue level organisation
Contractility	Ability to shorten on stimulation
Elasticity	Ability to return to original length
Smooth muscle	Tissue chiefly responsible for peristaltic waves in gut

(NCERT Ed. 2020-21, 11th, Page No. 104)

172. (3)

Compound epithelium is a type of epithelial tissue composed of multiple layers of cells. It's primarily found in areas that experience significant friction, abrasion, or stress, such as the skin, the lining of the mouth etc.

The basal layer (the deepest layer) of this tissue contains basal cells, which are undifferentiated stem cells. They are highly mitotic, meaning they divide frequently. This rapid division is essential for producing new cells that move upward through the layers of the epithelium to replace the older, superficial cells. These older cells are constantly being shed or damaged due to the wear and tear of daily activities. The continuous replacement of these cells maintains the integrity and protective function of the tissue.

(NCERT Ed. 2020-21, 11th, Page No. 102)

173. (4)

- Exocrine glands are characterized by the presence of ducts. These ducts are tubes that transport their secretions to a specific location, such as a body surface or a cavity. Examples of exocrine glands include salivary glands, sweat glands, and mammary glands. They produce a variety of substances like saliva, sweat, and digestive enzymes.
- Endocrine glands, on the other hand, are ductless. They release their secretions, which are specifically called hormones, directly into the surrounding interstitial fluid or bloodstream. The blood then carries these hormones to target cells and organs throughout the body. Examples include the thyroid gland, adrenal gland, and pituitary gland.

(NCERT Ed. 2020-21, 11th, Page No. 102)

174. (2)

Blood is classified as a specialized connective tissue. It meets the criteria because it:

- Connects different body systems by transporting substances like oxygen, nutrients, and hormones.
- Has a non-living liquid extracellular matrix called plasma.

- Originates from the mesoderm, one of the primary germ layers in embryonic development.

Fibrinogen is a soluble protein, not an active fibre. It's the precursor to fibrin, which forms the fibers of a blood clot. The fibres themselves (fibrin) are not normally circulating. Fibrinogen is not secreted by blood cells. It's a protein that's primarily produced and secreted into the bloodstream by the liver.

(NCERT Ed. 2020-21, 11th, Page No. 104)

175. (2)

A sheath of tough connective tissue encloses several bundles of muscle fibres which is known as the perimysium. This is a layer of dense, irregular connective tissue that surrounds a bundle of muscle fibers, known as a fascicle. The perimysium is an important part of the muscle structure, providing support and allowing for the passage of blood vessels and nerves.

(NCERT Ed. 2020-21, 11th, Page No. 104)

176. (3)

The animal kingdom (Kingdom Animalia) is defined by its multicellular organisms. Unicellular organisms, such as *amoebas* and *paramecium*, belong to other kingdoms.

Connective tissue is abundant and provides support, it doesn't "cover" all four tissue types. The tissue that forms linings and covers surfaces is epithelial tissue.

Adipose tissue is a type of specialised connective tissue that provides insulation. The fat stored in adipocytes acts as a barrier to heat loss, helping maintain body temperature. It protects organs by absorbing shock.

Skeletal tissues, which include bone and cartilage, form the skeletal system itself. The muscles that are attached to these bones are called skeletal muscles, but the skeletal tissues themselves are not named based on being attached to bones.

(NCERT Ed. 2025-26, 11th, Page No. 79)

177. (4)

Contraction in response to stimuli is the fundamental property of muscle tissue, enabling movement and force generation.

Muscle tone (partial contraction) maintains body posture without continuous conscious effort.

Skeletal muscles allow locomotion and quick responses to stimuli, helping in survival.

Muscle tissue is not limited to a few movements; it is involved in virtually all voluntary and many involuntary movements of the body. Skeletal muscles control voluntary movements, smooth muscles regulate internal organ movements, and cardiac muscle powers the heartbeat continuously.

(NCERT Ed. 2020-21, 11th, Page No. 104)

178. (3)

- A camel's hump is a large mass of adipose tissue, which is primarily composed of fat cells. This stored fat serves as a concentrated energy reserve. When the body oxidizes this fat for energy, a significant amount of water is produced as a byproduct. This metabolic water helps the camel survive for long periods without drinking, which is a crucial adaptation for its arid environment.
- Dense irregular connective tissue is present in skin.

(NCERT Ed. 2020-21, 11th, Page No. 103)

179. (2)

The description matches dense irregular connective tissue.

This type of connective tissue:

- Contains thick bundles of collagen fibres arranged in multiple directions, interwoven to provide tensile strength in many planes. Fibroblasts are the predominant cell type, producing collagen and extracellular matrix.
- Provides structural support and resists stretching forces from various directions.
- Commonly found in the dermis of the skin (beneath the epidermis) and also in fibrous capsules around joints and some organ coverings.

(NCERT Ed. 2020-21, 11th, Page No. 103)

180. (3)

Neuroglial cells are non-excitabile supporting cells of the nervous system. They vastly outnumber neurons and occupy more volume because they perform multiple supportive functions critical for neuron survival and efficient impulse conduction.

The main reason for their abundance is:

- They protect neurons by forming insulating barriers (e.g., myelin sheaths in Schwann cells and oligodendrocytes).
- They provide structural support, maintaining the architecture of the nervous system.
- They nourish neurons by supplying nutrients and maintaining the ionic and chemical environment necessary for nerve impulse transmission.
- They also remove metabolic waste and defend against pathogens via immune-like activities (e.g., microglia).

(NCERT Ed. 2020-21, 11th, Page No. 105)

